BCA I SEM MAIN EXAM ACD. SESSION 2025-26

BCA-104 - Computer Architecture

Time: 3 Hours **Maximum Marks: 70**

Instructions to Candidates:

- Part-A: Short answer questions (up to 25 words) 10x2 marks = 20 marks. All ten questions are compulsory.
- Part-B: Analytical/Problem Solving questions 5x4 marks = 20 marks. Candidates have to answer 5 questions out of 7.
- Part-C: Descriptive/Analytical/Problem Solving questions 3x10 marks = 30 marks. Candidates have to answer 3 questions out of 5.
- Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

PART-A

Q.1. Write 2's complement of binary number 10011011.

Q.2. Give truth table of EX-OR Gate.

Q.3. What are peripheral devices?

Q.4. Define addressing modes.

Q.5. Define Cache Memory.

Q.6. Differentiate between Computer organisation and Architecture.

Q.7. What is a Flip Flop?

Q.8. Give and brief a sample instruction format.

Q.9. What are modes of data transfer?

Q.10. Define memory Hierarchy.

PART-B

Q.1. With the help of a neat block diagram explain full adder circuit.

Q.2. Explain General Register organization of CPU in brief.

Q.3. Write a short technical note on I/O processor.

Q.4. What do you understand by memory mapped IO.

Q.5. Explain concept of paging in virtual memory.

Q.6. Draw block diagram of a decoder and explain it.

Q.7. Explain with example N's complement.

PART-C

Q.1. Write a detailed technical note on mapping techniques and replacement algorithms in cache memory.

Q.2. Explain DMA Controller with the help of suitable block diagram.

Q.3. Enumerate and explain various addressing modes.

Q.4. Define and differentiate between RISC and CISC computer architecture. Also give examples of instructions in each.

Q.5. Explain Strobe control and Handshaking in Asynchronous data transfer. Also brief First In First out buffer.